

# RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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## ABSTRACT

Audio temporal grounding maps a text query to the corresponding event intervals. When the same sounds recur in different orders, event recognition alone cannot determine which occurrence satisfies a relational query: multiple windows are acoustically valid, yet only one matches the requested relation. We introduce RelTwin, a fine-tuning framework that contrasts such locally valid timestamp answers. Shared sound excerpts form opposite-order windows, and inverse queries exchange their correct targets. RelTwin first learns query-dependent occurrence selection through complete-answer competition, then ranks interval sets by coverage, count, and boundary accuracy. Across SpotSound-Bench, Clotho-Moment, and UnAV-100, it improves SpotSound-A and TimeAudio by up to 4.23 and 14.48 mIoU points, respectively. On 160 inverse-query pairs, candidate supervision improves joint localization-and-selection accuracy by 31.25 points over matched paired SFT and reduces identical-answer pairs from 37 to 1. This advantage persists under altered silence timing, while inference retains direct timestamp generation.

**Index Terms**— Audio temporal grounding, timestamp answer contrasts, query–window binding, paired supervision

## 1. INTRODUCTION

Audio temporal grounding maps a natural-language query to event intervals in a recording [1, 2]. When sound categories repeat, recognition alone cannot determine which occurrence satisfies the query. A recording may contain both “dog followed by rooster” and “rooster followed by dog,” which inverse queries should map to different windows. Returning the same window for both is a *query–window binding error*: the prediction is acoustically supported but relationally incorrect.

Prior work advances several parts of audio grounding. Weakly supervised methods learn localization from clip-level

captions [3], while audio-language models support open-ended generation [4, 5, 6]. TimeAudio adds temporal markers and absolute time-aware encoding [7]; SpotSound interleaves timestamps with audio tokens and trains on absent-event queries [8]. These systems improve timestamp generation, but their positive sequence objectives do not explicitly compare two acoustically valid answers.

Compositional audio–language methods distinguish alternative descriptions through audio–text alignment [9], counterfactual captions [10], temporal contrast [11], composition-aware negatives [12], and multi-grain alignment [13]. SHINE constructs negative queries and saliency rankings for video grounding [14]. RelTwin instead scores complete timestamp answers for coexisting windows whose correctness exchanges with the query.

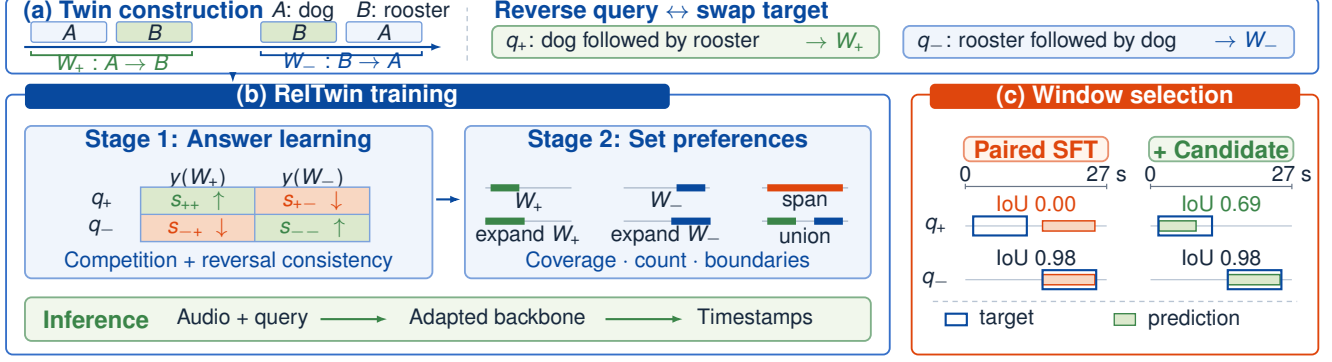
The challenge is to distinguish two *locally valid timestamp answers*, not to construct an obviously incorrect negative. Positive sequence supervision raises the correct answer’s likelihood without requiring it to outrank another acoustically supported window. A controlled contrast should hold the audio fixed, change the queried relation, and compare complete interval sets because errors can involve occurrence, coverage, count, or boundaries.

RelTwin (Fig. 1) introduces this competition through two-stage fine-tuning. Shared excerpts form opposite-order windows whose targets exchange under inverse queries. Stage 1 contrasts complete serialized answers and couples their preferences under reversal; Stage 2 compares interval sets with different coverage, counts, and boundaries. Timestamp-answer scores and localization replay adapt SpotSound-A and TimeAudio while preserving direct generation without candidate scoring at inference.

RelTwin formulates query–window binding as competition between complete timestamp answers for span-style and temporal-token outputs. Both backbones improve on three public benchmarks. In a matched Stage 1 comparison, candidate supervision raises JointPairAcc by 31.25 points and reduces identical answers from 37 to 1 of 160 inverse-query pairs. The advantage persists after changing silence timing while event waveforms, order, and queries remain fixed.

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**Fig. 1.** RelTwin. (a) Shared excerpts form opposite-order windows  $W_{\pm}$  for queries  $q_{\pm}$ . (b) Green/orange cells mark matching/competing timestamp answers; arrows indicate the desired score changes. Stage 2 compares interval sets. (c) SpotSound-A paired SFT versus candidate supervision: outlines are targets, filled bars are predictions, and numbers are set-IoU.

## 2. METHOD

RelTwin organizes supervision around competition between locally valid timestamp answers (Fig. 1). Section 2.1 constructs query-dependent twins from shared excerpts and inverse queries. Section 2.2 uses complete-answer competition (Stage 1) to select the queried window and reverse the preference with the query. Section 2.3 introduces quality-aware interval-set preferences (Stage 2) for coverage, count, and boundary errors. For recording  $x$  of duration  $T$  and query  $q$ , the output is  $S = \{[a_k, b_k]\}_{k=1}^n$ , where  $0 \leq a_k < b_k \leq T$ .

### 2.1. Query-dependent twin construction

This construction supplies the competing examples used by both training stages. To make the query determine window selection, we arrange the same ESC-50 excerpts [15],  $A$  and  $B$ , as  $A \rightarrow B$  and  $B \rightarrow A$  in one recording. Targets  $W_{\pm}$  span each sequence, including its gap; queries  $q_{\pm}$  request the respective orders. Both windows contain the same categories, but  $W_{-}$  is negative for  $q_{+}$  and positive for  $q_{-}$ . We alternate which order appears first.

### 2.2. Stage 1: complete-answer competition

Stage 1 establishes query-specific selection by comparing complete answers under each query. We serialize  $S$  as comma-separated spans for SpotSound-A or temporal tokens for TimeAudio. Answer tokens  $y(S)$  exclude the prompt and receive the mean log-probability

$$s_{\theta}(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_{\theta}(y_t | x, q, y_{<t}). \quad (1)$$

The  $2 \times 2$  matrix  $M_{ij} = s_{\theta}(x, q_i, W_j)$  indexes queries by rows and answers by columns,  $i, j \in \{+, -\}$  (Fig. 1b). Its diagonal entries are the correct query-answer pairs. Sequence supervision minimizes  $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$ , rewarding each correct answer independently. Candidate supervision adds the missing competition by training each diagonal

score to exceed the other acoustically valid answer in its row. For  $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$  with  $\tau > 0$ ,

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_{+}(+) + \log p_{-}(-)]. \quad (2)$$

Query reversal should also reverse the answer preference. We align the two distributions after exchanging candidate labels using the Jensen–Shannon divergence  $\mathcal{L}_{\text{JS}} = \text{JS}(p_{+}, \Pi p_{-})$ , where  $\Pi$  swaps the candidates. Replay retains ordinary localization supervision through examples  $(x_r, q_r, S_r)$  with  $\mathcal{R}_1 = -s_{\theta}(x_r, q_r, S_r)$ . Stage 1 optimizes

$$\mathcal{L}_1 = \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{cand}} + \gamma \mathcal{L}_{\text{JS}}) + \beta \mathcal{R}_1. \quad (3)$$

### 2.3. Stage 2: quality-aware interval-set preferences

Stage 2 assigns graded preferences to imperfect answers according to coverage, interval count, and boundaries, rather than treating them as equally wrong. Its six candidates are  $W_{\pm}$ , their expanded versions, their enclosing span, and their union. Expansions expose boundary excess; the span includes intervening background; the union adds the competing occurrence.

For candidate  $S_j$  and target  $W_i$ , a four-axis quality vector  $h_{ij}$  measures set-IoU, threshold margin, count agreement, and boundary accuracy. The threshold axis averages soft F1 at  $\eta = 0.3/0.5$ . Each interval’s best-match IoU  $u$  contributes  $\max(0, u - \eta)/(1 - \eta)$ ; averaging over predictions and targets gives soft precision and recall. With  $n_S$  denoting interval count, count agreement is  $1 - |n_{S_j} - n_{W_i}| / \max(n_{S_j}, n_{W_i}, 1)$ . Boundary accuracy is  $\exp(-e)$ :  $e$  averages the summed endpoint error for each target’s best-IoU match, divided by  $2 \max(b - a, 0.1)$  for target  $[a, b]$ .

Among  $K$  candidates, let  $d_{ij}^{+}$  and  $d_{ij}^{-}$  count those dominated by and dominating  $S_j$ . Dominance means no worse on any axis and better on at least one. We define utility as

$$v_{ij} = \frac{1}{2} \left[ \frac{1}{4} \sum_{\ell=1}^4 h_{ij\ell} + \frac{d_{ij}^{+} - d_{ij}^{-}}{K - 1} \right]. \quad (4)$$

Mean quality preserves graded differences; dominance rewards improvements without sacrificing another axis. Soft targets  $t_i = \text{softmax}(v_i/\tau_q)$  assign higher probability to better interval sets. The same answer scores, normalized over all six candidates, learn this distribution through  $\mathcal{L}_{\text{set}} = -\frac{1}{2} \sum_{i,j} t_i(j) \log p_i(j)$ . The six-way permutation  $\Pi = (2, 1, 4, 3, 5, 6)$  swaps targets and expansions, fixing the span and union.

Replay candidates perturb boundaries, add/drop events, merge spans, split the longest interval, or cover the recording. The same quality rule defines their listwise loss  $\mathcal{L}_{\text{set},r}$ . On replay examples, the frozen base model supplies a reference distribution  $p_0$ ; the KL constraint anchors preferences  $p_r$  to this reference to help retain existing localization preferences. Stage 2 optimizes

$$\begin{aligned}\mathcal{L}_2 &= \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{set}} + \gamma\mathcal{L}_{\text{JS}}) + \beta\mathcal{R}_2, \\ \mathcal{R}_2 &= \mathcal{R}_1 + \lambda\mathcal{L}_{\text{set},r} + \kappa \text{KL}(p_0||p_r).\end{aligned}\quad (5)$$

Both backbones share Eq. 1. Training updates LoRA parameters and, for TimeAudio, temporal/audio connectors. Inference remains direct timestamp generation from  $(x, q)$ , without twin construction or candidate scoring.

### 3. EXPERIMENTS

#### 3.1. Experimental setup

*Datasets.* We evaluate general localization on SpotSound-Bench (400 queries, 387 audios) [8], Clotho-Moment (6,649 queries), and the UnAV-100 subset (100 queries, 77 audios) [2]. For controlled relation learning, 40 ESC-50 classes produce 256 training recordings, 1,024 relation queries (512 pairs), and 512 category queries. The templates “followed by” and “and then” give two inverse-query pairs per recording. Relation development uses 10 held-out classes, 80 recordings, and 320 queries (160 pairs), with classes and original Freesound IDs disjoint from adaptation data.

*Evaluation.* Public benchmarks use set-mIoU and R1 at IoU 0.3/0.5, with predictions clipped to the recording duration. For inverse-query pairs, let  $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$ , where  $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$  and an empty union scores zero. PairAcc requires  $\min(u_{++}, u_{--}) \geq 0.5$ ; a swap occurs if  $u_{++} \leq u_{+-}$  or  $u_{--} \leq u_{-+}$ , including ties. Joint-PairAcc requires PairAcc without a swap. Inspired by paired compositional evaluation [16], it separates selection from coverage: the union of two disjoint targets can pass PairAcc but fails JointPairAcc. These are pair-level percentages; same-answer counts require identical interval lists.

*Training and implementation.* The complete RelTwin configuration uses 384 Stage 1 and 128 Stage 2 updates for both backbones. LoRA [17] and bridge learning rates are  $2 \times 10^{-6}$  and  $5 \times 10^{-6}$ ; exchange/JS weight is 1, replay weight 0.75, and reference-KL weight 0.2. TimeAudio and SpotSound-A use 31.76M and 37.63M trainable parameters,

respectively; both use seed 0 and load adapter+bridge weights at inference. We use  $\lambda = \gamma = 1$ ,  $\beta = 0.75$ ,  $\kappa = 0.2$ ,  $\tau = 1$ ,  $\tau_q = 0.15$ , and 15% interval expansion with a 0.05 s minimum. Decoding is greedy with one beam and at most 128 new tokens.

Table 2 isolates Stage 1 competition on SpotSound-A. Paired SFT uses  $\lambda = 0$  and candidate supervision  $\lambda = 1$ , both with  $\gamma = 0$  and  $\beta = 0.5$ . They share initialization, data, order, and 256 AdamW [19] updates at  $5 \times 10^{-6}$ , with zero weight decay and gradient clipping 1. LoRA has rank 8,  $\alpha = 16$ , dropout 0.1, and 20.19M trainable parameters. Each update accumulates one inverse-query pair and one replay query. Candidate supervision adds four gradient-free answer scores and weighted gradient recomputation to SFT’s two positive-answer backpropagations. All variants use 16 kHz mono audio, bfloat16, and direct timestamp generation.

For the timing intervention, we recompose 80 development recordings with a 1.5 s lead-in, 0.25 s within-window gaps, 3 s between windows, and a 2 s tail while preserving waveforms, order, and queries. Frozen matched models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain identical.

TimeAudio and matched point estimates use seed 0; Table 2(c) reports candidate supervision over seeds 0/1/2 as mean  $\pm$  sample SD. We obtain 95% CIs from 20,000 audio-group bootstrap resamples; relation tests resample 39 connected source-sharing groups (largest: 18 recordings).

#### 3.2. Grounding performance across benchmarks

RelTwin improves both backbones on all three benchmarks (Table 1). TimeAudio gains 8.16, 5.48, and 14.48 mIoU points; SpotSound-A gains 1.53, 1.08, and 4.23. R1 rises at both thresholds on every dataset, and adapted SpotSound-A leads every listed column. Gains with span-style and temporal-token outputs support complete-answer supervision across timestamp representations.

#### 3.3. Controlled ablation: complete-answer competition

Under matched data and updates, paired SFT rewards both correct answers independently, whereas candidate supervision requires each to outrank the competing window. Candidate supervision improves SpotSound mIoU by 1.02 points over paired SFT (95% audio-group CI: [0.23, 1.81]; Table 2).

JointPairAcc rises by 31.25 points (source-group CI [22.35, 40.50]). Identical-answer pairs fall from 37 to 1 out of 160; 27 of those 37 become jointly correct. Figure 1 illustrates this recovery. Across all pairs, transitions to joint correctness outnumber regressions by 51 to 1.

The relation mIoU gain is 13.97 points (CI [9.84, 17.51]). Because both candidate windows contain the queried categories, producing distinct, jointly correct answers demonstrates selection of the queried relation rather than category detection alone.

**Table 1.** Temporal grounding across three benchmarks with RelTwin adaptation of TimeAudio and SpotSound-A (%; higher is better). Baseline results: SpotSound [8, Table 3].  $\Delta$ : gain over the corresponding backbone. Bold marks column maxima.

| Method                                           | SpotSound-Bench |              |              | Clotho-Moment |              |              | UnAV-100 subset |              |              |
|--------------------------------------------------|-----------------|--------------|--------------|---------------|--------------|--------------|-----------------|--------------|--------------|
|                                                  | R1@.3           | R1@.5        | mIoU         | R1@.3         | R1@.5        | mIoU         | R1@.3           | R1@.5        | mIoU         |
| <i>Non-LLM models</i>                            |                 |              |              |               |              |              |                 |              |              |
| WTATG                                            | 53.0            | 34.5         | 38.4         | 12.1          | 6.3          | 9.1          | 53.0            | 37.0         | 38.4         |
| AM-DETR                                          | 25.5            | 15.0         | 19.5         | 89.8          | 88.0         | 80.9         | 59.0            | 46.0         | 42.8         |
| <i>Proprietary models</i>                        |                 |              |              |               |              |              |                 |              |              |
| Gemini-2.5-Flash                                 | 34.8            | 27.8         | 25.7         | 45.7          | 33.7         | 36.9         | 48.0            | 37.0         | 35.6         |
| Gemini-2.5-Pro                                   | 20.0            | 17.3         | 19.7         | 40.4          | 32.5         | 32.5         | 39.0            | 35.0         | 34.6         |
| <i>Open-source models and RelTwin adaptation</i> |                 |              |              |               |              |              |                 |              |              |
| Kimi-Audio                                       | 3.0             | 1.3          | 3.2          | 0.7           | 0.1          | 0.9          | 8.0             | 2.0          | 5.3          |
| Qwen2-Audio                                      | 5.5             | 1.8          | 4.4          | 6.1           | 1.3          | 5.7          | 14.0            | 4.0          | 9.7          |
| Audio Flamingo 3                                 | 8.8             | 2.8          | 7.6          | 32.9          | 21.8         | 22.6         | 35.0            | 24.0         | 25.0         |
| TimeAudio                                        | 7.8             | 1.8          | 10.2         | 39.5          | 24.9         | 28.6         | 21.0            | 7.0          | 16.0         |
| TimeAudio + RelTwin                              | 24.25           | 13.25        | 18.36        | 45.63         | 30.92        | 34.08        | 44.00           | 25.00        | 30.48        |
| $\Delta$ vs. TimeAudio                           | +16.45          | +11.45       | +8.16        | +6.13         | +6.02        | +5.48        | +23.00          | +18.00       | +14.48       |
| SpotSound-A                                      | 76.5            | 59.5         | 57.9         | 93.4          | 91.0         | 85.6         | 86.0            | 74.0         | 69.8         |
| SpotSound-A + RelTwin                            | <b>78.00</b>    | <b>62.00</b> | <b>59.43</b> | <b>93.61</b>  | <b>91.01</b> | <b>86.68</b> | <b>89.00</b>    | <b>79.00</b> | <b>74.03</b> |
| $\Delta$ vs. SpotSound-A                         | +1.50           | +2.50        | +1.53        | +0.21         | +0.01        | +1.08        | +3.00           | +5.00        | +4.23        |

**Table 2.** Mechanism experiment: isolating RelTwin’s candidate supervision on SpotSound-A. (b) Matched data and updates; (c) mean  $\pm$  sample SD. Joint: JointPairAcc; same-answer: inverse-query pairs with identical timestamp answers. Scores are percentages except counts.

| Method                                                    | SpotSound        | Relation development (160 inverse-query pairs) |                  |                         |                                |
|-----------------------------------------------------------|------------------|------------------------------------------------|------------------|-------------------------|--------------------------------|
|                                                           | mIoU $\uparrow$  | mIoU $\uparrow$                                | Joint $\uparrow$ | Swap error $\downarrow$ | Same-answer pairs $\downarrow$ |
| <i>(a) Backbone reference: shared evaluation protocol</i> |                  |                                                |                  |                         |                                |
| SpotSound-A                                               | 58.42            | 37.64                                          | 0.63             | 65.00                   | 15/160                         |
| <i>(b) Matched training comparison</i>                    |                  |                                                |                  |                         |                                |
| Paired SFT                                                | 58.42            | 74.58                                          | 53.13            | 40.00                   | 37/160                         |
| Candidate supervision                                     | <b>59.43</b>     | <b>88.55</b>                                   | <b>84.38</b>     | <b>7.50</b>             | <b>1/160</b>                   |
| Gain over paired SFT                                      | +1.02            | +13.97                                         | +31.25           | −32.50                  | −36 pairs                      |
| <i>(c) Candidate supervision: three training seeds</i>    |                  |                                                |                  |                         |                                |
| Candidate supervision                                     | 59.16 $\pm$ 0.26 | 87.05 $\pm$ 1.43                               | 81.46 $\pm$ 2.53 | 8.13 $\pm$ 0.63         | (1, 0, 0)/160                  |

### 3.4. Binding gains persist after timing intervention

On the 40 recordings with altered silence timing, candidate supervision retains a 20-point JointPairAcc advantage (87.50 vs. 67.50; CI [2.94, 33.33]), compared with original scores of 85.00 and 42.50. The advantage persists when event waveforms, order, and queries are fixed but silence timing changes.

Fixed silence duration or window placement therefore cannot solely explain the binding gain. Public benchmarks test general localization. The matched ablation isolates complete-answer competition, while temporal recomposition tests whether relation selection survives changed timing. Stage 2 extends this competition from occurrence choice to interval coverage, count, and boundaries.

## 4. CONCLUSION

RelTwin addresses query–window binding by contrasting acoustically valid timestamp answers with different relational meanings. It improves two backbones on three benchmarks, selects different correct windows for inverse queries, and retains the advantage after silence timing changes. The framework learns relation-aware occurrence selection while preserving direct timestamp generation.

Answer competition links sound categories to query-specific instance selection.

## 5. ACKNOWLEDGEMENTS

**Compliance with Ethical Standards.** We use public ESC-50 [15], SpotSound [8], Clotho-Moment, UnAV [2], TEMPO, and AudioGrounding data, including programmatically constructed sequences. The study involved no new recordings, participant recruitment, human or animal experiments, or listening studies.

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